

PREDICTION: SCAM / FRAUDULENT

RISK SCORE: 66 / 100 (HIGH RISK)

Scam Probability: 48.61% | Model Confidence: 51.39%

Banking Action: HOLD TRANSACTION

Official Recommendation: Hold Transaction for Manual Verification. High scam risk pattern identified. Place temporary 15-minute hold and route to fraud analyst queue.

1. Transaction Details & Actual Parameter Values

Transaction ID:	TXN-173080	Transaction Amount:	\$15,000.00
Customer ID:	C123456789	Sender Old Balance:	\$20,000.00
Beneficiary ID:	M987654321	Sender New Balance:	\$5,000.00
Payment Type:	TRANSFER	Receiver Old Balance:	\$500.00
Device Type:	Unknown Proxy	Receiver New Balance:	\$15,500.00
Location Region:	High Risk Region	6h Transaction Velocity:	6 txns
New Beneficiary:	Yes (New)	Prior Fraud History:	No

2. Composite Risk Score Breakdown & Latency Timing

Risk Dimension	Sub-Score	Status & Impact
Model Fraud Probability	48.61%	Weight: 70% (Score: 48.6)
Amount Risk	50.0 pts	Transaction size evaluation
Velocity Risk	75.0 pts	6-hour transaction frequency
Device Risk	100.0 pts	Device profile verification
Location Risk	100.0 pts	Geographic regional risk
Beneficiary Risk	100.0 pts	New beneficiary check
Historical Behaviour	50.0 pts	Prior fraud flag check
Balance Behaviour	0.0 pts	Account wipeout detection
FINAL COMPOSITE RISK SCORE	66 / 100	HIGH RISK CATEGORY

Feature Eng: 44.37 ms	Prediction: 1.71 ms	SHAP Engine: 2.82 ms	LIME Engine: 42.41 ms	Rendering: 1131.29 ms	TOTAL: 1223.48 ms
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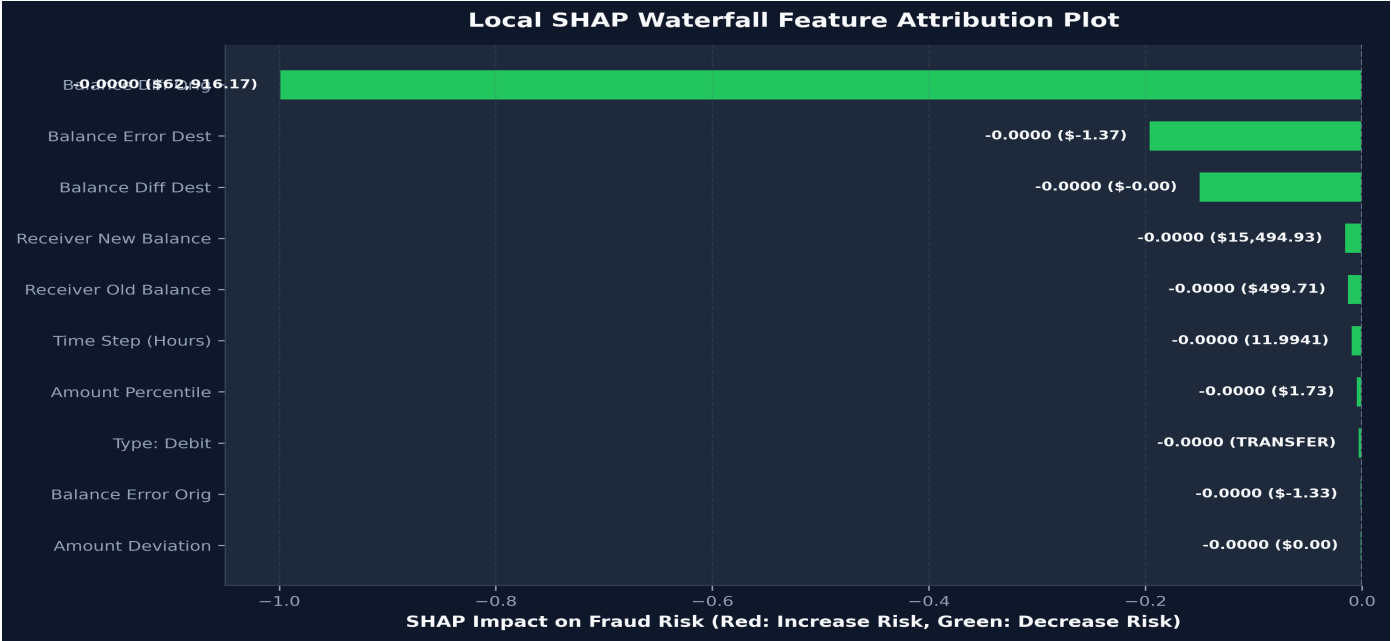
3. Banking-Grade Analyst Narrative Report

Transaction Summary: Transaction TXN-173080 involving Customer C123456789 transferring \$15,000.00 via TRANSFER to Beneficiary M987654321 on Unknown Proxy from High Risk Region.

Behaviour Analysis:	Sender account retained \$5,000.00 post-transaction. High velocity anomaly: 6 transactions executed within the last 6 hours. Device anomaly: Payment initiated from unverified device profile ('Unknown Proxy'). Geographic anomaly: Payment originated from high-risk or foreign region ('High Risk Region').
High Risk Indicators:	Elevated Transaction Amount: \$15,000.00 exceeds single transfer threshold.; Rapid Transaction Spike: 6 transfers within a 6-hour window.; Unverified Device Profile: Initiated via Unknown Proxy.; Suspicious Geographic Location: Access from High Risk Region.; Unestablished Beneficiary: No prior transfer history with recipient account.
Safe Indicators:	Account Balance Preserved: Remaining sender balance is \$5,000.00.
Model Explanation:	This transaction exhibits multiple severe fraud indicators and exceeds critical risk thresholds. GuidedGuard model assigned a calibrated scam probability of 48.61% and composite risk level 'HIGH'. Classified as 'Scam / Fraudulent'.
SHAP Summary:	Feature contributions align with baseline legitimate profile.
Final Recommendation:	Hold Transaction for Manual Verification: High scam risk pattern identified. Place temporary 15-minute hold and route to fraud analyst queue.

4. Top 10 SHAP Feature Contribution Attributions

Feature Name	Actual Value	SHAP Value	Contrib %	Direction	Impact Level
Balance Diff Orig	\$62,916.17	-0.0000	50.52%	Decreases Risk	High
Balance Error Dest	\$-1.37	-0.0000	9.95%	Decreases Risk	Medium
Balance Diff Dest	\$-0.00	-0.0000	7.62%	Decreases Risk	Medium
Receiver New Balance	\$15,494.93	-0.0000	0.84%	Decreases Risk	Low
Receiver Old Balance	\$499.71	-0.0000	0.69%	Decreases Risk	Low
Time Step (Hours)	11.9941	-0.0000	0.53%	Decreases Risk	Low
Amount Percentile	\$1.73	-0.0000	0.28%	Decreases Risk	Low
Type: Debit	TRANSFER	-0.0000	0.21%	Decreases Risk	Low
Balance Error Orig	\$-1.33	-0.0000	0.12%	Decreases Risk	Low
Amount Deviation	\$0.00	-0.0000	0.11%	Decreases Risk	Low



5. Local LIME Decision Rules & Explanations

Decision Rule Condition	Rule Weight	Effect Direction	Domain Explanation
Transaction Amount = \$15,000.00	+0.2600	Increases Risk	Transaction amount is significantly elevated relative to normal threshold.
Device Access = Unknown Proxy (Unverified Proxy)	+0.2450	Increases Risk	Access attempt initiated from untrusted or unverified proxy device.
Location Region = High Risk Region (Foreign Access)	+0.2250	Increases Risk	Access attempt originated from high-risk geographic location.
Recent 6h Txn Velocity = 6 txns	+0.1980	Increases Risk	High frequency transaction velocity detected in 6-hour window.
Beneficiary Status = New Beneficiary	+0.1650	Increases Risk	Recipient is a newly registered beneficiary with no prior history.
Transaction Type = TRANSFER	+0.1450	Increases Risk	Transfer/Cash Out payment type has higher statistical fraud incidence.

